

MARABA, Brazil

WMO: 825620

Lat: 5.3661S

Lon: 49.1252W

Elev: 102

StdP: 100.11

Time Zone: -3.00 (W03)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	20.0	21.0	15.9	11.4	33.6	17.0	12.3	31.3	5.5	26.3	4.8	26.4	0.7	200	0.317

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	12.3	38.1	25.2	37.1	25.1	36.2	25.2	28.4	32.9	27.8	32.5	27.4	32.2	2.8	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.3	30.4	26.9	22.8	30.0	26.2	21.9	29.1	92.1	33.3	89.7	32.9	87.9	32.1	35.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.1	4.3	3.9	DB	18.4	39.2	1.0	1.6	17.7	40.4	17.1	41.3	16.5	42.2	15.8	43.4	(4)
(5)			WB	17.4	30.1	0.8	2.2	16.8	31.7	16.3	33.0	15.9	34.2	15.3	35.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.6	27.7	27.6	28.0	28.3	28.5	28.5	28.5	29.3	30.0	29.7	28.9	28.2	(6)	
	DBStd	1.39	1.37	1.31	1.20	1.20	1.02	0.94	0.99	1.10	1.22	1.46	1.29	1.26	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6794	550	494	559	550	574	556	574	597	599	610	567	564	(10)	
	CDD18.3	3752	292	261	301	300	316	306	316	339	349	352	317	306	(11)	
	CDH23.3	38791	2617	2271	2727	2833	3131	3198	3439	3955	4154	4077	3429	2959	(12)	
(13) Wind	CDH26.7	18338	998	847	1028	1111	1386	1705	1969	2261	2230	2040	1551	1211	(13)	
	WSAvg	1.5	1.5	1.5	1.5	1.4	1.2	1.3	1.4	1.7	2.0	1.9	1.5	1.4	(14)	
	PrecAvg	1843	300	318	370	257	96	29	17	12	43	89	142	231	(15)	
	PrecMax	2389	571	614	617	463	212	78	107	45	102	219	343	520	(16)	
	PrecMin	1389	58	98	104	35	6	0	0	0	5	5	45	126	(17)	
(18) Precipitation	PrecStd	312	126	141	110	115	55	23	24	13	28	56	71	90	(18)	
	DB	35.1	35.1	35.2	36.1	36.1	37.2	38.8	39.2	39.9	39.1	36.8	35.9		(19)	
	MCWB	27.0	27.0	27.4	27.5	27.4	25.5	23.8	24.9	24.6	26.0	26.6	27.5		(20)	
	DB	33.9	33.9	34.0	34.2	35.0	36.1	37.0	37.9	38.0	37.0	35.2	34.1		(21)	
	MCWB	26.9	26.9	27.2	27.3	26.9	25.7	24.4	24.5	25.4	26.3	26.5	26.7		(22)	
(23) Mean Coincident Wet Bulb Temperatures	DB	32.8	32.7	32.9	33.1	33.9	35.0	36.0	36.8	36.9	35.8	34.2	33.1		(23)	
	MCWB	26.7	26.7	27.0	26.9	26.8	25.4	24.3	24.3	25.2	26.0	26.4	26.4		(24)	
	DB	31.8	31.2	31.8	32.0	32.9	34.0	35.0	35.9	35.8	34.8	33.2	32.2		(25)	
	MCWB	26.4	26.5	26.6	26.6	26.4	25.5	24.4	24.2	25.2	25.9	26.2	26.2		(26)	
	WB	28.5	28.6	28.6	28.8	29.1	28.8	27.1	27.1	27.7	28.1	28.2	28.7		(27)	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	31.9	32.0	33.0	33.1	33.4	34.6	33.1	34.8	34.1	33.2	33.3	32.3		(28)	
	WB	27.5	27.5	27.7	27.9	27.7	27.2	26.5	26.4	26.7	27.1	27.4	27.5		(29)	
	MCDB	31.6	31.8	32.0	32.4	32.5	32.7	32.9	34.1	33.8	33.5	32.8	31.9		(30)	
	WB	27.0	26.9	27.2	27.4	27.2	26.6	25.9	26.0	26.2	26.6	27.0	27.0		(31)	
	MCDB	31.1	31.1	31.3	31.6	31.8	32.1	32.5	33.5	33.5	33.0	32.1	31.4		(32)	
(33) Mean Daily Temperature Range	WB	26.5	26.5	26.7	26.9	26.7	26.2	25.4	25.5	25.8	26.2	26.6	26.5		(33)	
	MCDB	30.2	30.1	30.4	30.8	31.1	31.6	31.8	32.5	33.0	32.3	31.3	30.6		(34)	
	MDBR	8.4	8.3	8.5	8.8	9.7	12.2	13.9	14.2	12.3	10.7	9.4	8.8		(35)	
	MCDBR	10.0	10.0	9.9	10.3	11.1	13.8	15.6	15.9	14.1	12.3	10.8	10.1		(36)	
	MCWBR	4.0	4.0	3.8	3.9	4.0	4.6	4.7	4.7	4.0	3.3	3.4	3.7		(37)	
(42) Clear-Sky Solar Irradiance	MCDBR	9.0	9.1	9.0	9.4	10.0	11.5	12.9	13.4	12.3	10.9	9.8	9.4		(38)	
	MCWBR	4.0	4.0	3.8	3.9	4.0	4.4	4.5	4.4	3.8	3.6	3.5	3.8		(39)	
	taub	0.400	0.394	0.391	0.392	0.402	0.381	0.390	0.423	0.519	0.553	0.528	0.466		(40)	
	taud	2.491	2.521	2.536	2.529	2.480	2.527	2.472	2.340	2.074	2.004	2.081	2.282		(41)	
(43) All-Sky Solar Radiation	Ebn at Noon	937	941	931	905	869	874	871	864	807	792	815	871		(42)	
	Edh at Noon	115	112	109	106	106	99	106	126	170	185	172	141		(43)	
(44) All-Sky Solar Radiation	RadAvg	4.61	4.70	4.80	4.82	5.00	5.31	5.58	5.78	5.30	4.88	4.64	4.55		(44)	
	RadStd	0.35	0.35	0.16	0.20	0.26	0.22	0.25	0.22	0.32	0.28	0.22	0.21		(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page